

MATTER

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EXAMINED AT THE TINIEST SCALE, the universe's matter is composed of elementary particles, some of which, governed by various forces, group together to form atoms and ions. In addition to these well-understood types of matter, other forms exist. Most of the universe's mass consists of this "dark matter," whose exact nature is still unknown.

WHAT IS MATTER?

Matter is anything that possesses mass—that is, anything affected by gravity. Most matter on Earth is made of atoms and ions. Elsewhere in the universe, however, matter exists under a vast range of conditions and takes a variety of forms, from thin interstellar medium (see p.228) to the matter in neutron stars and black holes (see p.267). Not all of this matter is made of atoms, but all matter is made of some kind of particle. Certain types of particles are elementary—that is, they are not made of smaller subunits. The elementary particles within ordinary matter are quarks and electrons, which make up atoms and ions. Most of the universe's matter, however, is not ordinary matter, but dark matter (see p.27), perhaps composed partly of neutrinos, theoretical WIMPs (weakly interacting massive particles), or both.

LUMINOUS MATTER

These illuminated gas clouds in interstellar space are made of ordinary matter, composed of atoms and ions.



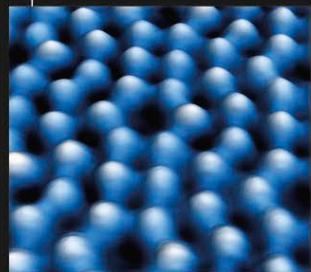
EMPTY SPACE
Most of an atom is empty—the protons, neutrons, and electrons are all shown here much larger than their real size relative to the whole atom.

STRUCTURE OF A CARBON ATOM

At the center of the atom is the nucleus, which contains protons and neutrons. Electrons move around within two regions, called shells, surrounding the nucleus. The shells appear fuzzy because electrons do not move in defined paths.

OUTER ELECTRON SHELL
Region in which four electrons orbit

INNER ELECTRON SHELL
Region within which two electrons orbit



IMAGING ATOMS

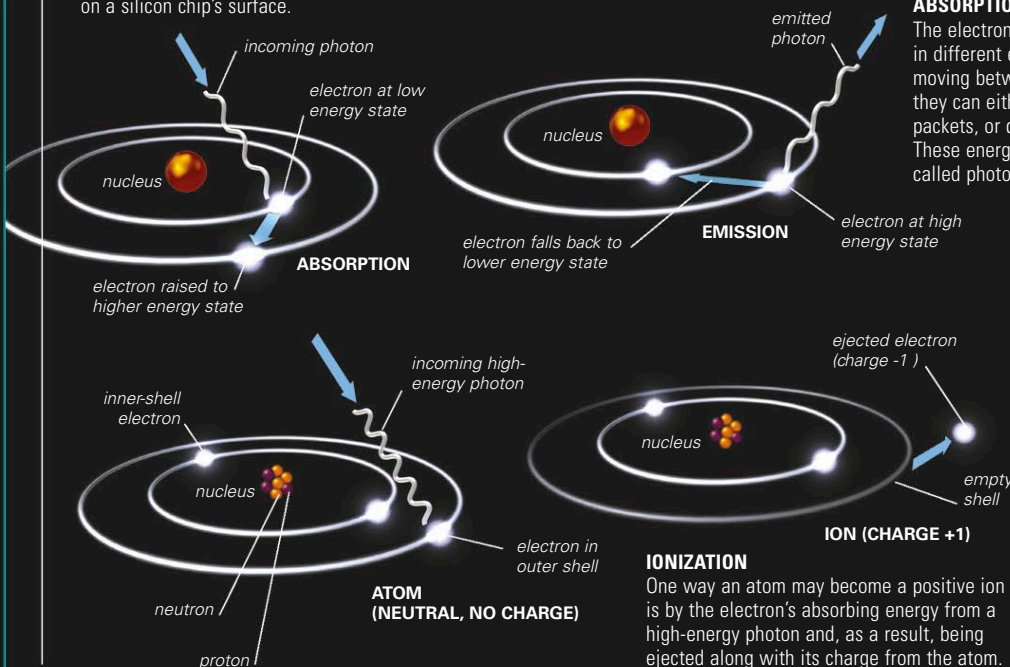
A device called a scanning-tunneling microscope was used to image these atoms on a silicon chip's surface.

ATOMS AND IONS

The quarks and electrons that make up atoms occur within specific regions of each atom. The quarks are bound in groups of three by gluons, which are massless particles of force. The quark groups form particles called protons and neutrons. These are clustered in a compact region at the center of the atom called the nucleus. Most of the rest of an atom is empty space, but moving around within this space are electrons. These carry a negative electrical charge and have a very low mass—nearly all the mass in an atom is in the protons and neutrons. Atoms always contain equal numbers of protons (positively charged) and electrons (negatively charged) and so are electrically neutral. If they lose or gain electrons, they become charged particles called ions.

ABSORPTION AND EMISSION

The electrons in atoms can exist in different energy states. By moving between energy states, they can either absorb or emit packets, or quanta, of energy. These energy packets are called photons.



red quark

gluon

green quark

blue quark

neutron

proton